

SGP-VSD: PACKAGER FOR SIP PACKETS ORIGINATING FROM THE DIGITALIZATION OF VIDEO DOCUMENTS IN HEALTHCARE

VideoSaúde Preservation Management System • VSD / ICICT / Fiocruz

A preservation workflow integrating BRAVS, archival description, preservation metadata and OAIS-aligned information packages.



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CORE IDEA | SGP-VSD structures the preservation dossier from the moment an audiovisual item is selected, connecting collection metadata, digitization control, archival description, automated technical metadata and BagIt packaging.

SYSTEM FOUNDATION

ARCHITECTURE & STACK

The SGP-VSD (VideoSaúde Preservation Management System) was developed by Lab-VSD/ICICT/Fiocruz, built upon a robust and scalable architecture.

Infrastructure & Persistence:

Isolated Docker containers with persistent volumes for storing data, raw video, and preservation dossiers, ensuring high portability and security.



TECH STACK

PHP ≥ 8.0.2
Laravel 9.x
MySQL 8.x
Docker Multi-stage
Nginx + Supervisor
FFmpeg • MediaInfo
Vite + Tailwind CSS

BRAVS INTEGRATION



BRAVS (Health Audiovisual Resource Bank) manages the collection (4,155 items: U-matic, VHS, Betacam, MiniDV, DVCam, etc.).

Full Sync via REST API consumes 20+ institutional endpoints.

Local mirrored tables protect selected cataloguing metadata from later overwriting.

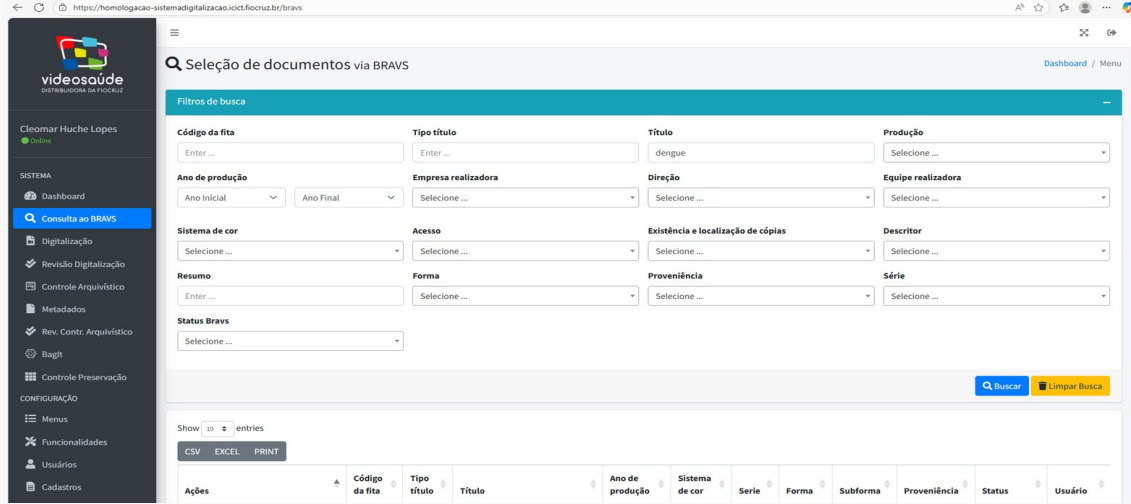
PRESERVATION WORKFLOW

Operational workflow detailed in sequential steps, featuring rigorous quality control and regulatory compliance.

Dossier/SIP Creation: Media selection initiates the dossier and automatically generates the folder structure and standard archival notation.

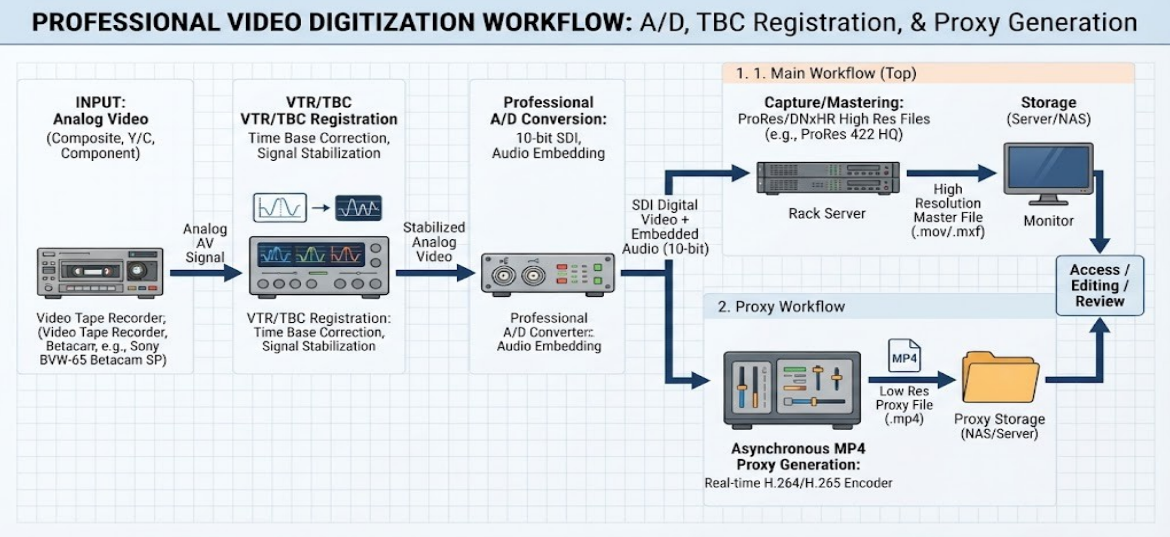
01 CONSULT & SELECT BRAVS

Collection search and dossier initiation with standardized notation.



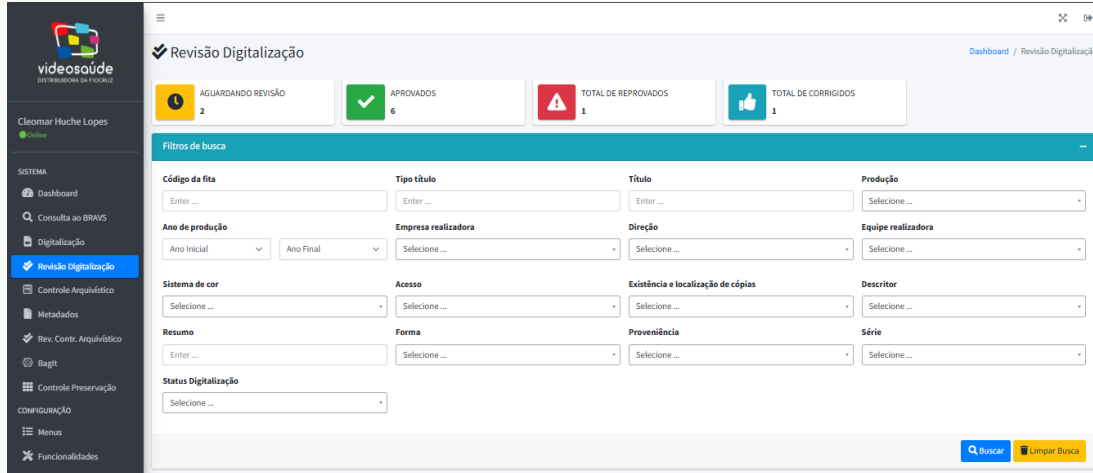
02 DIGITIZE & CAPTURE AV

Professional A/D conversion, VTR/TBC registration and asynchronous MP4 proxy generation.



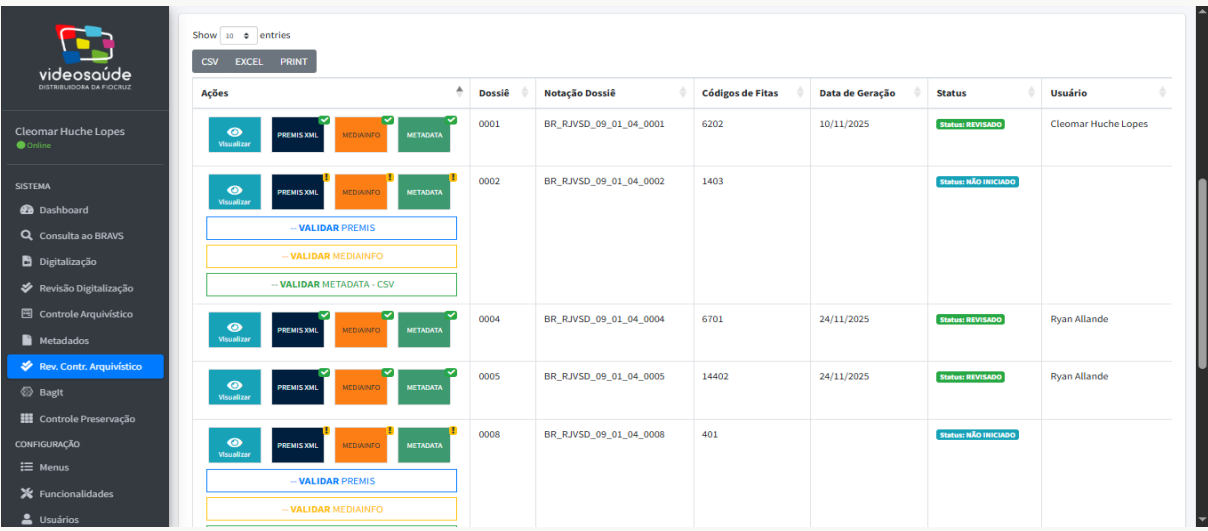
03 REVIEW QUALITY

Independent reviewer validates digitization as approved or rejected.



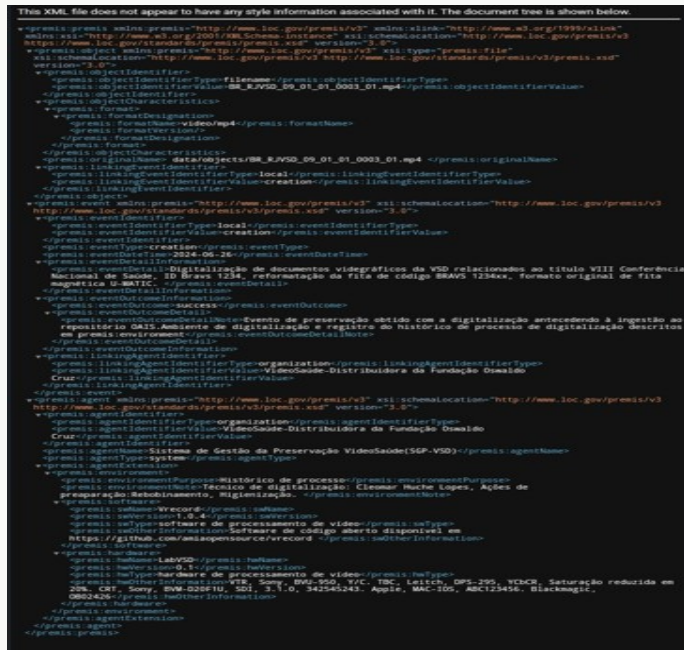
04 ARCHIVAL CONTROL DESCRIPTION

Archival description according to NOBRADE, with Dublin Core and EAD.



05 GENERATE METADATA AUTOMATION

PBCore2 extraction (via MediaInfo) + PREMIS v3.0 XML (Object, Event (capture/preparation actions), and Agent (environment/hardware/software/operator) + Dublin Core + EAD (metadata.csv).

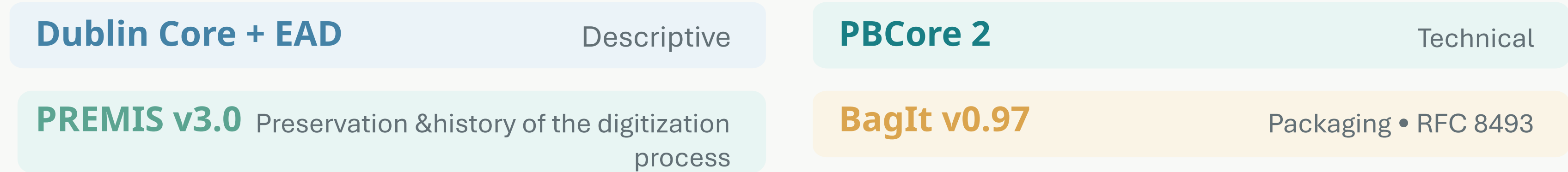


06 VALIDATE & PACKAGE BAGIT

Triple validation and BagIt packaging with UTF-8 manifests and MD5 hashes.

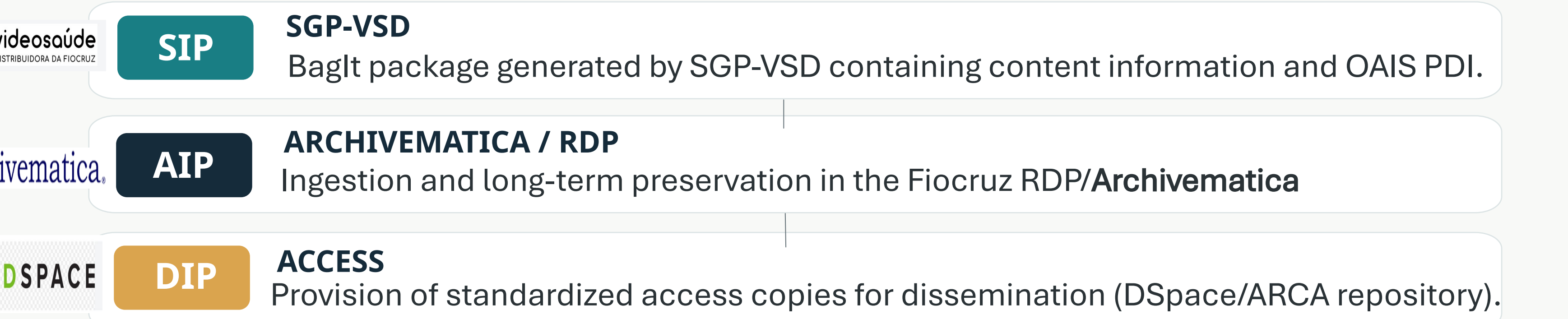


STANDARDS & METADATA



OAIS-ALIGNED INFORMATION PACKAGE FLOW

The SGP-VSD operates as a generator of SIP packages aligned with the OAIS reference model (ISO 14721:2012) for integration with the Fiocruz Digital Preservation Repository (RDP):



KEY TAKEAWAY | The SGP-VSD connects collection management, digitization control, archival description and preservation metadata into a traceable workflow capable of producing OAIS-aligned submission packages for institutional digital preservation.